Take home Midterm Problems and HW 4 Assignment 7502 Prof. Justin Frantz

Due Thursday March 26, 2026

Part 1 Take-Home Midterm Problems [Possibly still ~one more of these parts to be added, but nothing of the current problems will change except for typo fixes/clarifications] Please work on these independently, although you can always ask help from Prof. Frantz

M1) In a collider experiment, a decay of a certain heavy baryon can be identified from the decay to two particles, π^- and a proton as these particles are seen to emanate from a displaced vertex in a tracking detector (vertex located displaced from the beam/target locations). In a particular decay, the momenta of the π^- and *p* are measured to be 0.75 GeV and 4.25GeV respectively, and the opening angle between the tracks is 9 deg. Use the masses of the pion and proton of 139.6MeV and 938.3 MeV.

a) Calculate the mass of the baryon.

b) On average, over many such observed decays with the same total parent momentum, the displaced vertex is on average 0.35m from the point of production. Calculate the lifetime of the baryon.

M2) Two beams collide in an asymmetric energy collider ie a particle *a* in one beam and a particle *b* in the other have different energies. The collider intersects the beams such that (meaning the actual collisions occur when this is true) the standard convention that both beam *a* and *b* can be taken to be moving along the z-axis. In both parts below, the hyperbolic function relations involving *y* may be helpful –for example and also refer to rapidity problem 4 from HW3.

a) Show that the rapidity of the center of mass frame y_{CM} can be calculated by finding the rapidity of the four vector $(p_a^\mu + p_b^\mu)$. One way to do this is write the relations to explicitly transform p_z for *a* and *b* to the center of mass, and impose the required relation between p_a and p_b there.

b) Use this result to show that if a and b are same mass particles, that $y_{CM} = \frac{1}{2}(y_a + y_b)$.

c) Also use the basic transformation property of y given on the slides (first derived relation/property there) to show that in the CM frame the two particles have rapidities $\pm\frac{1}{2}(y_a - y_b)$

M3) Similar to the example in Griffiths regarding the ratio of cross sections for N+N → d + π, use isospin eigenstate kets to motivate why it is expected that the total cross section ratios for σ_{p+p}, σ_{p+n} , & (and σ_{n+n} _) (e.g. pp/np) are all one, ie all cross sections should be considered equal for elastic scattering in some energy regime, like high but not too high energy.

Part 2 Regular Homework 4 Portion [Definitely still more to be added, though no problems will change]

H1) Using the expression for dσ/d *t* for 1+2-> 3+4 given in the slides, (and using our natural units throughout)

a) Express the formula using only Mandelstam variables if $m_1 = m_2 = m$. (& other constants but for all kine factors... ie instead of |pi*| or |pf*|)

H2) Show that $\bar{u}(p_1)\gamma^\mu u(p_1) = K\, p_1^\mu$ where K is a constant

H3) Partially Based on Peskin Problem 3.2

The two QFT reference books (Peskin et al/Lancaster et al), probably most QFT books, and some particle physics books apparently prefer a different ("alt") convention than the one Griffiths uses (apparently NOT Bjorken and Drell's per Griffiths) for the γ matrices explicit forms/representation. The difference is simply that the "alt" convention (apparently also called the Chiral or Weyl convention) "switch" γ^0 and -γ^5 . Meaning the negative of the traceless matrix Griffiths calls γ^5 is the one they call γ^0 and the one Griffiths calls γ^0 they call γ^5. This allows for amongst other things a more compact notation since e.g. then all components of γ^μ including μ = 0 have 2x2 block form with zero [0_{2x2}] as the diagonal blocks and off-diags written in terms of σ^μ = (1_{2x2} ,$\vec{\sigma}$).

a) Compare the explicit form (in terms of p^μ) of the matrix $\gamma^\mu p_\mu$ = $\not{p}$ between the two conventions.

b) Compare the explicit form of the matrix $\sigma^{\mu\nu}$ defined in equation 7.69 in Griffiths for $\mu = 0$ and ν = 1 between the two conventions.

c) [Peskin 3.2] Prove the "Gordon Identity":

$$\bar{u}(p')\gamma^\mu u(p) = \bar{u}(p')\left(\frac{p'^\mu + p^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p'_\nu - p_\nu)}{2m}\right)u(p).$$

Start from the RHS and simplify till you obtain the LHS using anti/commutation relations, the Dirac equation and possibly γ matrix identities given in Griffiths such as those in the previous problem (Giff problem 7.32). You may presume these all still apply as given in Griffiths, even in the case of the alt convention, for the purposes of this problem. For sure the anti-commutator = $2g^{\mu\nu}$ relation MUST still hold for the Dirac trick of factorizing the Klein-Gordon equation to work.

PS.: In my solution, the explicit forms of the γ matrices were not needed, and therefore which convention shouldn't matter for this part of the problem. But just in case I had an error, if you get stuck you may want to verify whether any of the Griffiths relations you used still apply exactly as given when using the alt convention-- this is not required, only a suggestion if you run into trouble. Also, note, in the alt convention, the free particle solution u(p) choice is also generally defined differently and/or there may be a slightly different normalization for *u(p)* in the alt choice (or even perhaps just because of a further choice of norm particular to Peskin & Schroeder) but try not to use the explicit forms for the u(p)'s to do this problem, you shouldn't need them, only that $\psi = u(p)e^{ip\cdot x}$ is the form of the solution to the Dirac equation. In case normalization conventions are different, you need only prove the relation up to an overall constant.